

Application 2

Optimal portfolios

expected return

Fund (AA)

$E(r)$

15%

σ

20%

return
volatility
weights
or
st. dev.

80%

T-Bills

4%

0%

20%

Portfolio P : 80% in AA
20% in T-Bills

Portfolio Stats

$$E(r_p) = 80\% \times 15\% + 20\% \times 4\%$$

Expected return

Varience:

$$\sigma_p^2 = (20\%)^2 \times (80\%)^2 + (0\%)^2 \times (20\%)^2 + 2 \times \sigma_{AA, T-Bills} \times 80\% \times 20\%$$

Covariance of AA & T-Bills



T-Bills has no risk

$$\sigma_p^2 = (20\%)^2 \times (80\%)^2$$

$$\sigma_p = \sqrt{\sigma_p^2} = 20\% \times 80\%$$

More generally

If we replace 80% by y ,
the portfolio with weights
 y in AA & $1-y$ in T-Bills
has stats:

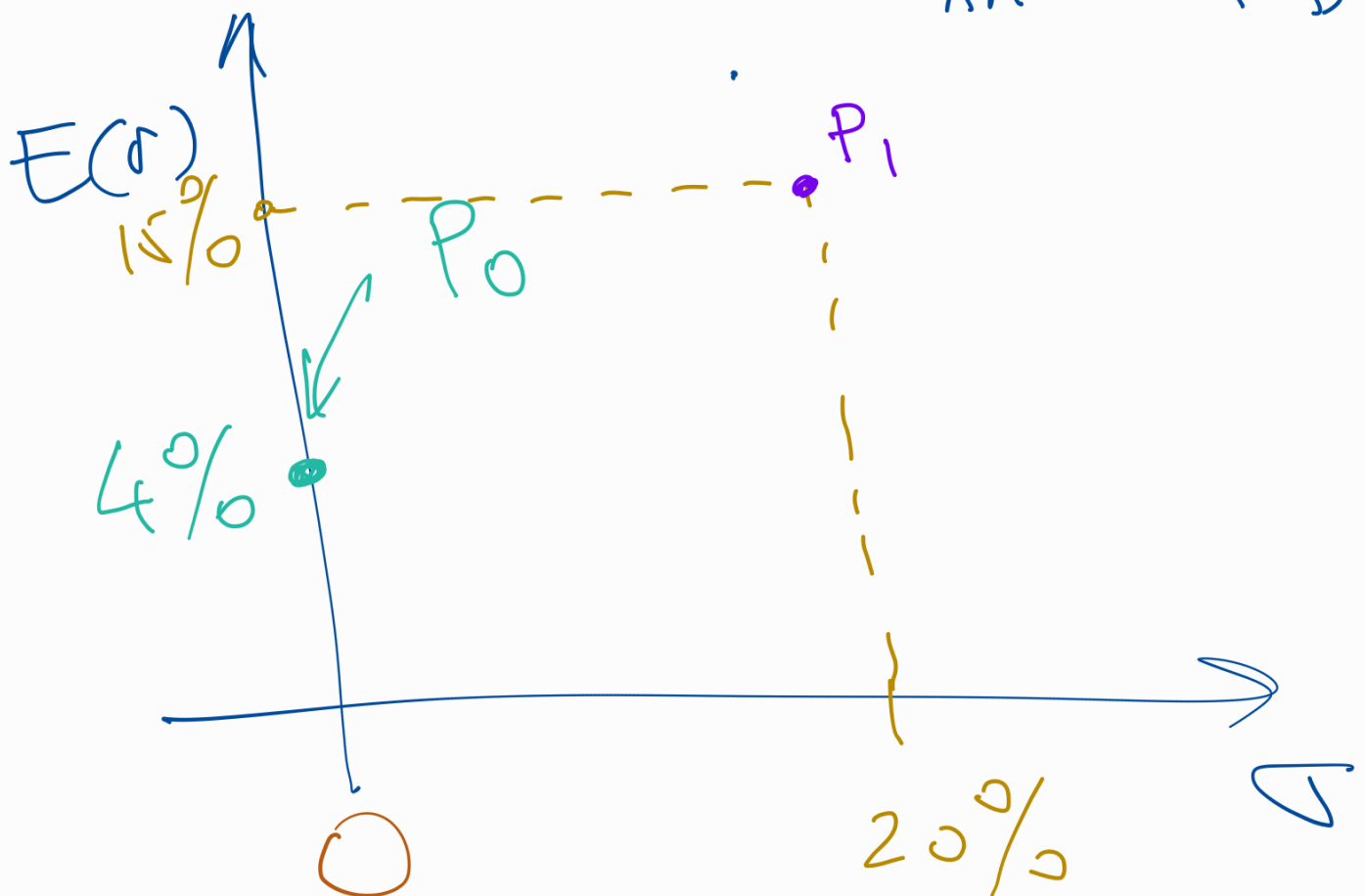
$$E(r_p) = y \times 15\% + (1-y) \times 4\%$$

$$\sigma_p = y \times 20\%$$

Let's plot $E(r_p)$ against σ_p for several values of y

$$y = 0 \rightarrow P_0 = (0\%, 100\%)$$

$\uparrow$ AA $\uparrow$ T-Bills



$$y = 1 \Rightarrow P_1 = (100\%, 0\%)$$

$\uparrow$ AA $\uparrow$ T-Bills

How do we select between P_0 & P_1 ?

Like higher return
Dislike risk

Happiness Function :
(Utility)

$$U_p = E(r_p) - \frac{1}{2} \boxed{A} \sigma_p^2$$

Risk aversion

We can now calculate :

$$U(P_0) < U(P_1)$$

?

We choose the portfolio
(i.e., y) that generates
highest happiness



Optimal portfolio